

Software Architecture Assignment 2: Quality Attributes and Quality Metrics

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1. Introduction and Case Summary

Selected System

The selected system is a Geolocation-Based Service Worker Marketplace Platform.

System Overview

The platform connects customers who require immediate or scheduled services (such as plumbing, electrical work, cleaning, and maintenance) with nearby service workers using real-time geolocation. Customers can post service requests, discover and compare nearby workers, communicate in-app, track job progress, and complete secure payments. Service workers can manage profiles, browse and bid on jobs, receive real-time notifications, and track earnings. Administrators oversee verification, disputes, analytics, and system health.

The platform is designed as a scalable, highly available microservices-based system to support rapid growth, variable demand, and strict security and compliance requirements.

2. Reference to Assignment 1 Non-Functional Requirements

This assignment builds directly on the non-functional requirements defined in Assignment 1:

- **Performance:** Search response time < 2 seconds, real-time geolocation updates within 500 ms, support for 10,000+ concurrent users
- **Scalability:** Horizontal scaling to support 500% user growth and independent microservice scaling
- **Security:** End-to-end encryption, OAuth 2.0 authentication, GDPR and PCI-DSS compliance
- **Availability & Reliability:** 99.9% uptime SLA, automated failover, regular backups
- **Usability:** Mobile-responsive design, WCAG 2.1 compliance, multi-language support
- **Maintainability:** Independent deployment, automated testing > 80%, monitoring and alerting

Assumptions and constraints identified in Assignment 1 (e.g., microservices architecture, third-party payment gateways, highly variable traffic patterns) remain unchanged.

3. Selected Quality Attributes (ISO/IEC 25010)

Based on the system context and stakeholder priorities, the following ISO/IEC 25010 quality attributes are architecturally significant:

1. **Performance Efficiency** – Critical for fast search, booking, and real-time interactions
2. **Reliability** – Required to ensure continuous service availability and fault tolerance
3. **Security** – Essential due to financial transactions and sensitive personal data
4. **Usability** – Key to customer adoption and service worker engagement
5. **Maintainability** – Necessary for rapid feature evolution and operational efficiency

Other attributes such as Portability and Compatibility are considered secondary and not prioritized at this stage.

4. Quality Requirements and Metrics

4.1 Performance Efficiency

Description

Performance efficiency refers to the system's ability to provide fast and responsive behavior under expected and peak workloads.

Refined Quality Requirements - Service search and booking must feel instantaneous to end users - Real-time updates must not degrade system responsiveness during peak load

Quality Metrics - Average search response time ≤ 1.5 seconds under normal load - 95th percentile search latency ≤ 2 seconds during peak load - Geolocation update propagation time ≤ 500 ms - Support $\geq 10,000$ concurrent active users without SLA violation

4.2 Reliability

Description

Reliability measures the system's ability to operate continuously and recover quickly from failures.

Refined Quality Requirements - The platform must remain available despite partial service failures - Data loss must be minimized during outages

Quality Metrics - Monthly availability $\geq 99.9\%$ - Mean Time To Recovery (MTTR) ≤ 15 minutes - Mean Time Between Failures (MTBF) ≥ 30 days - Recovery Point Objective (RPO) ≤ 1 hour

4.3 Security

Description

Security ensures protection of user data, financial transactions, and system access against unauthorized actions.

Refined Quality Requirements - Only authenticated and authorized users can access protected resources - Sensitive data must be protected in transit and at rest

Quality Metrics - 100% of API traffic encrypted using TLS - Zero critical security vulnerabilities in production - Number of security incidents ≤ 1 per quarter - Payment services fully compliant with PCI-DSS audits

4.4 Usability

Description

Usability reflects how easily customers and service workers can achieve their goals using the platform.

Refined Quality Requirements - Core user tasks should require minimal effort and learning - The platform must be accessible to users with disabilities

Quality Metrics - Task completion rate $\geq 95\%$ for core flows (booking, bidding) - Average number of steps to complete booking ≤ 3 - WCAG 2.1 AA compliance score = 100% - User-reported usability issues ≤ 5 per month

4.5 Maintainability

Description

Maintainability represents the ease with which the system can be modified, tested, and extended.

Refined Quality Requirements - Changes to one service should not require system-wide redeployment - Defects should be detected early through automated testing

Quality Metrics - Automated test coverage $\geq 80\%$ - Average deployment time per service ≤ 10 minutes - Change failure rate $\leq 10\%$ - Mean time to implement minor feature ≤ 2 weeks

5. Quality Attribute Prioritization

Quality Attribute	Priority	Justification
Performance Efficiency	High	Directly impacts customer satisfaction and marketplace success
Reliability	High	Service downtime leads to lost revenue and trust
Security	High	Financial data and regulatory compliance are critical
Usability	Medium	Important for adoption but can tolerate gradual improvement
Maintainability	Medium	Enables long-term evolution but less visible to end users

Lower-priority attributes such as Portability may be partially degraded if required to achieve performance and reliability goals.

6. Architectural Tactics and Design Decisions

Performance Efficiency Tactics

- Caching of geolocation and search results to reduce database load
- Asynchronous processing for notifications and analytics
- Load balancing and horizontal scaling of high-demand services

Reliability Tactics

- Redundant service instances across multiple availability zones
- Health checks and automated failover mechanisms
- Graceful degradation of non-critical features during overload

Security Tactics

- Centralized authentication and authorization mechanisms
- Encryption of all sensitive data in transit and at rest
- Audit logging and monitoring of security-related events

Maintainability Tactics

- Modular microservices with well-defined APIs
- Continuous integration and automated testing pipelines
- Clear separation of concerns between services

7. Trade-offs and Impact Analysis

1. Performance vs. Maintainability

Aggressive caching and optimized data flows improve performance but increase system complexity, making maintenance more challenging. This trade-off is acceptable due to strong performance requirements.

2. Security vs. Usability

Strong authentication and session management introduce additional user steps but are justified to protect sensitive data and payments.

3. Availability vs. Cost Efficiency

High availability through redundancy increases infrastructure costs, but this is acceptable given business-critical uptime requirements.

8. Risks and Measurement Strategy

Key Risks - Unpredictable traffic spikes may violate performance metrics - Misconfigured security controls could expose sensitive data - Insufficient monitoring could delay failure detection

Measurement and Monitoring - Continuous monitoring of latency, error rates, and availability - Security audits and vulnerability scans on a regular basis - User analytics and feedback collection for usability assessment

Quality metrics are continuously evaluated and refined throughout the system lifecycle.

9. Assumptions and Open Questions

Assumptions - Third-party payment gateways remain available and compliant - Users prioritize speed over absolute geolocation accuracy - Microservices architecture remains the chosen structural style

Open Questions - What level of performance degradation is acceptable during extreme peak loads? - How will future regulatory changes impact security and data retention requirements? - Should premium users receive differentiated quality guarantees?

Conclusion

This assignment demonstrates how high-level non-functional requirements from Assignment 1 are translated into ISO/IEC 25010 quality attributes, measurable metrics, and justified architectural decisions. The resulting analysis provides a clear foundation for evaluating and evolving the system architecture based on quality-driven reasoning.